Ordinary Differential Equations

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October 15, 2026

Contents

1 Nonhomogeneous Linear Equations **2**

1.1 Choosing a Trial Particular Solution 2

1.2 Example: Resonance with a Trigonometric Forcing 2

1.3 Example: A Repeated Root and Polynomial Forcing 3

1.4 Superposition for Several Forcing Terms 3

1.5 Example: Three Forcing Terms 4

1.6 When to Use a Different Method 4

1 Nonhomogeneous Linear Equations

A second-order nonhomogeneous linear differential equation has the form

$$L[y] := y'' + p(t)y' + q(t)y = g(t).$$

The associated homogeneous equation is $L[y] = 0$. If y_h is its general solution and y_p is any one solution of $L[y_p] = g$, then

$$\boxed{y = y_h + y_p}$$

is the general solution of the nonhomogeneous equation. Indeed, linearity gives $L[y_h + y_p] = L[y_h] + L[y_p] = g$; conversely, the difference of any two solutions of $L[y] = g$ solves $L[y] = 0$.

For constant coefficients, the homogeneous part comes from the characteristic equation. The **method of undetermined coefficients** then chooses a trial form for y_p whose derivatives have the same types of terms as g. It applies when g is a finite sum of polynomials, exponentials, sines, cosines, and their products. Once the trial is chosen, substitution determines its unknown coefficients.

1.1 Choosing a Trial Particular Solution

For a polynomial $P_n(t)$ of degree n, use a general polynomial $Q_n(t)$ of the same degree. Typical trial forms are

Forcing term	Initial trial for y_p
$P_n(t)$	$Q_n(t)$
$e^{\alpha t}P_n(t)$	$e^{\alpha t}Q_n(t)$
$P_n(t)\cos\beta t$ or $P_n(t)\sin\beta t$	$Q_n(t)\cos\beta t + R_n(t)\sin\beta t$
$e^{\alpha t}P_n(t)\cos\beta t$ or $e^{\alpha t}P_n(t)\sin\beta t$	$e^{\alpha t}\big(Q_n(t)\cos\beta t + R_n(t)\sin\beta t\big)$

Include both sine and cosine terms because differentiation mixes them. Likewise, include every polynomial power through degree n.

Resonance. If any term in the trial solves the homogeneous equation, the trial cannot produce the forcing term. Multiply the entire trial by t^s, where s is the multiplicity of the corresponding characteristic root $\alpha + i\beta$. Then substitute and solve for the coefficients. For real coefficients, complex roots come in conjugate pairs and share the same multiplicity.

1.2 Example: Resonance with a Trigonometric Forcing

Solve

$$y'' + 4y = 3\cos 2t.$$

Solution.

The characteristic equation $r^2 + 4 = 0$ has roots $\pm 2i$, so

$$y_h = C_1\cos 2t + C_2\sin 2t.$$

The usual trial $A\cos 2t + B\sin 2t$ is already homogeneous and therefore gives zero when substituted into $y'' + 4y$. The roots have multiplicity one, so use

$$y_p = t(A\sin 2t + B\cos 2t).$$

Direct differentiation shows

$$(D^2+4)(t\sin 2t) = 4\cos 2t, \qquad (D^2+4)(t\cos 2t) = -4\sin 2t.$$

Thus $4A = 3$ and $B = 0$. The general solution is

$$\boxed{y = C_1\cos 2t + C_2\sin 2t + \frac{3}{4}t\sin 2t.}$$

1.3 Example: A Repeated Root and Polynomial Forcing

Solve

$$y'' - 4y' + 4y = te^{2t}.$$

Solution.

The characteristic polynomial is $r^2 - 4r + 4 = (r-2)^2$. Hence

$$y_h = (C_1 + C_2 t)e^{2t}.$$

The forcing is an exponential times a degree-one polynomial, and $r = 2$ is a double root. Multiply the ordinary trial by t^2:

$$y_p = t^2 e^{2t}(At + B) = e^{2t}(At^3 + Bt^2).$$

For any smooth v,

$$(D-2)^2\big(e^{2t}v(t)\big) = e^{2t}v''(t).$$

With $v = At^3 + Bt^2$, the equation becomes

$$e^{2t}(6At + 2B) = te^{2t}.$$

Therefore $A = 1/6$ and $B = 0$, giving

$$\boxed{y = e^{2t}\left(C_1 + C_2 t + \frac{t^3}{6}\right).}$$

1.4 Superposition for Several Forcing Terms

If $g = g_1 + \cdots + g_m$ and $L[y_{p,j}] = g_j$, then

$$L[y_{p,1} + \cdots + y_{p,m}] = g_1 + \cdots + g_m.$$

Find one particular solution for each forcing term and add them. This works because L is linear. After adding y_h, combine or absorb any duplicate homogeneous terms into its constants.

1.5 Example: Three Forcing Terms

Find a particular solution of the equation shown in the class notes:

$$y'' - 3y' - 4y = 3e^t + 2\sin t - 8e^t \cos 2t.$$

Solution.

Write $L = D^2 - 3D - 4$. Split the right-hand side into three terms.

For $3e^t$, try $y_{p,1} = Ae^t$. Since $L[e^t] = (1 - 3 - 4)e^t = -6e^t$, we obtain

$$y_{p,1} = -\frac{1}{2}e^t.$$

For $2\sin t$, try $y_{p,2} = A\sin t + B\cos t$. Comparing coefficients gives

$$-5A + 3B = 2, \qquad -3A - 5B = 0,$$

so $A = -5/17$ and $B = 3/17$. Thus

$$y_{p,2} = -\frac{5}{17}\sin t + \frac{3}{17}\cos t.$$

For $-8e^t \cos 2t$, try $y_{p,3} = e^t(A\cos 2t + B\sin 2t)$. Substitution gives

$$-10A - 2B = -8, \qquad 2A - 10B = 0,$$

so $A = 10/13$ and $B = 2/13$. Therefore

$$\boxed{y_p = -\frac{1}{2}e^t - \frac{5}{17}\sin t + \frac{3}{17}\cos t + \frac{10}{13}e^t \cos 2t + \frac{2}{13}e^t \sin 2t.}$$

The homogeneous characteristic equation $(r - 4)(r + 1) = 0$ gives $y_h = C_1 e^{4t} + C_2 e^{-t}$. Hence the full general solution is

$$\boxed{y = C_1 e^{4t} + C_2 e^{-t} + y_p.}$$

1.6 When to Use a Different Method

Undetermined coefficients relies on the forcing term staying in a finite-dimensional family under differentiation. A forcing term such as $\tan t$, $\csc t$, or $\ln t$ generally requires another method. **Variation of parameters** handles such terms when a fundamental pair of homogeneous solutions is known.